

SHIMIN LUO

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EDUCATION

Northwestern University

Ph.D. in Computer Science

- **Advisor:** [Prof. Jennie Rogers](#)

Evanston, IL, USA

Sep. 2024 – Present

Southern University of Science and Technology (SUSTech)

BS in Computer Science and Technology

- **GPA:** 3.86/4.00, **Rank:** 11/220 (5%)
- Research Assistant at **Database Research Group**
- **Advisor:** [Prof. Bo Tang](#) (SUSTech), [Prof. Huan Li](#) (Zhejiang University)

Shenzhen, China

Sep. 2020 – Jun. 2024

University of Illinois Urbana-Champaign

Summer Internship

- **Advisor:** [Prof. Kevin C.C. Chang](#)
- **Research Topic:** Graph Data Mining on Science Analytics

Champaign-Urbana, IL, USA

Jun. 2023 – Aug. 2023

University of California, Berkeley

Visiting student, Major: Computer Science

Berkeley, CA, USA

Aug. 2023 – Dec. 2023

INTERESTS

- Privacy-Preserving Database, Oblivious Query Processing, Database Performance Modeling
- Data Management, Private Data Federations
- Cloud-Native Database, Cloud-Edge Computing, Distributed Computing

RESEARCH EXPERIENCE

COMPASS: Cost-Optimal MPC-as-a-service for Secure Analytical Queries

Northwestern

Keywords: Cost-Optimality, MPC-as-a-Service, Query Performance Prediction

- Aims to build the first framework to achieve cost-optimality for secure analytical queries in the cloud. By leveraging Query Performance Prediction (QPP) and multi-level parallel orchestration, COMPASS automates resource management while maintaining strong privacy guarantees for data and access patterns, effectively minimizing the economic cost of secure computations.

SODE: Proximity-based Streaming Outlier Detection in Edge Computing Environments

SUSTech

Keywords: Streaming Outlier Detection, Edge Computing, Decentralized Distributed Computing Framework

- **System Architecture Innovation:** Designed and implemented a novel decentralized distributed framework for real-time outlier detection in streaming data. Architected a 5-step collaborative edge-device interaction protocol and 2 efficient pruning rules to maximize detection efficiency and ensure data security.
- **Algorithm & Optimization:** Pioneered a specialized MCOD variant for edge computing by introducing the "underfilled clusters" concept and optimizing corresponding algorithm to enhance outlier detection. Generalized and integrated two state-of-the-art algorithms (MCOD&NETS) into the framework.
- **High-Performance Implementation:** Implemented the entire system using Apache Thrift (RPC) for efficient decentralized communication; Achieved 5x faster detection than P2P and 2x higher efficiency than centralized baselines.
- **System Scalability:** Developed the framework to support heterogeneous multi-query workloads, allowing concurrent, device-specific outlier metrics across distributed edge nodes.

Graph Data Mining on Science Analytics

UIUC

Keywords: Graph Data Mining, Information Retrieval & Prediction, Entity Matching

- **Graph Engine & Data Quality:** Developed a data pipeline to model OpenAlex scholarly data into Neo4j; performed rigorous cross-source validation against DBLP and Google Scholar to identify and resolve large-scale metadata inconsistencies.
- **Multimodal Profiling & Innovation:** Architected a profiling engine using name/image heuristics for demographic inference; proposed a novel temporal-based proxy to analyze academic trends across age gradients and generations.
- **Algorithmic Analysis:** Optimized the MVR algorithm for superior accuracy; designed an end-to-end analytical framework to investigate longitudinal performance gaps across gender and ethnicity in academia.
- [Github repository](#) | Preparation Part: [Report](#), [MVR demo video](#); Summer Research: [Report](#), [Code demo video](#)

COMPETITION

SIGMOD 2023 Programming Contest, World Finalist

May, 2023

- The goal is to build an approximate K-NN Graph for 10 million, 100-dimension vectors.
- The main approach involves adopting the NN-descent concept, optimizing the computational process (e.g., using SIMD) and the choice of data structures (using heap and sort-array in a phased manner) to reduce complexity.

OUTSTANDING PROJECTS

End-to-end Secure File Sharing System (Leader)

UCB, 2023 Fall

- Developed a secure system using cryptographic libraries for user authentication, file storage and sharing against attacks.
- Role: Designer, developer and tester.
- Ranked 12th out of 315, enhanced expertise in **system design and data security**, and honing problem-solving skills.

Online Course Platform (Leader)

SUSTech, 2022 Fall

- An online course platform with various modules: video, practice, homework, payment, notice, etc.
- Role: Backend developer, as well as database constructor and manager.
- Accumulated experience in **data management** and **full-stack project development**.

P2P File Transfer on UDP With Congestion Control (Leader)

SUSTech, 2022 Fall

- Developed a reliable P2P file transfer system using UDP, featuring handshaking, reliability, concurrency, congestion control, and peer crash recovery.
- Role: Designer, developer, tester.
- Gained experience in **socket and parallel distributed programming**, improving overall system understanding.

Database Component Implementation (Independent)

Northwestern, 2025 Spring

- Implemented database storage engine, B+ tree index and various operators in RustDB.
- Gained experience in **database implementation** and **Rust programming**.

Database Application and Data Visualization (Independent)

SUSTech, Fall 2022

- Extracted and stored GitHub repository data; analyzed (e.g., contributions, preferred working time) and visualized.
- Accumulated experience in **data crawling, cleaning, analysis, and visualization**.

AWARDS AND HONORS

- Outstanding Graduate (three times, across University, Department & Shuren College) 2024
- 1st Prize, University Merit Student Scholarship (top 5%) 2021
- 2nd Prize, University Merit Student Scholarship (twice, top 10%) 2022 & 2023
- Shuren College Scholarship 2021
- Exemplary Student Leader 2022

TEACHING EXPERIENCE

Teaching Assistant - Introduction to Database

2026 Spring

Teaching Assistant - Computer Organization

2023 Spring

Teaching Assistant - Data Structure and Algorithm Analysis

2022 Fall

Responsibilities:

- Assist the instructor in preparing assignments and class discussion
- Design the course projects and provide demos for students
- Hold office hour and grade students' assignments & labs

DEPARTMENT & STUDENT SERVICES

- Member, Graduate Women in Computing (GWiC) | Northwestern University
- Volunteer, CS PhD Visiting Day 2025 | Northwestern University
- Alumni Representative & Coordinator | SUSTech Class of 2024
- Class Representative, Department of Computer Science and Engineering | SUSTech Sep. 2022 - July 2024
- Student Union Department Administrator and Financial Administrator | SUSTech Sep. 2020 - Oct. 2022

COMPUTER AND LANGUAGE SKILLS

- **Programming Languages:** Java, Python, C/C++, SQL, Rust, Go, Cypher, JavaScript, Verilog, MATLAB, Scheme
- **Technology Stack:** Vue3, SpringBoot, MyBatis, PostgreSQL, Neo4j, Node.js, Junit, JavaFX, Swing, Docker
- **Developer Tools:** Git, VS Code, PyCharm, IntelliJ, CLion, DataGrip, WebStorm, Vivado, Maven, Swagger, Anaconda, Jupyter Notebook